

Ouroboros Quantum Convexity Engine V6.0

Institutional Multi-Asset Quantitative Strategy Whitepaper (2016–2026 Out-of-Sample Audit)

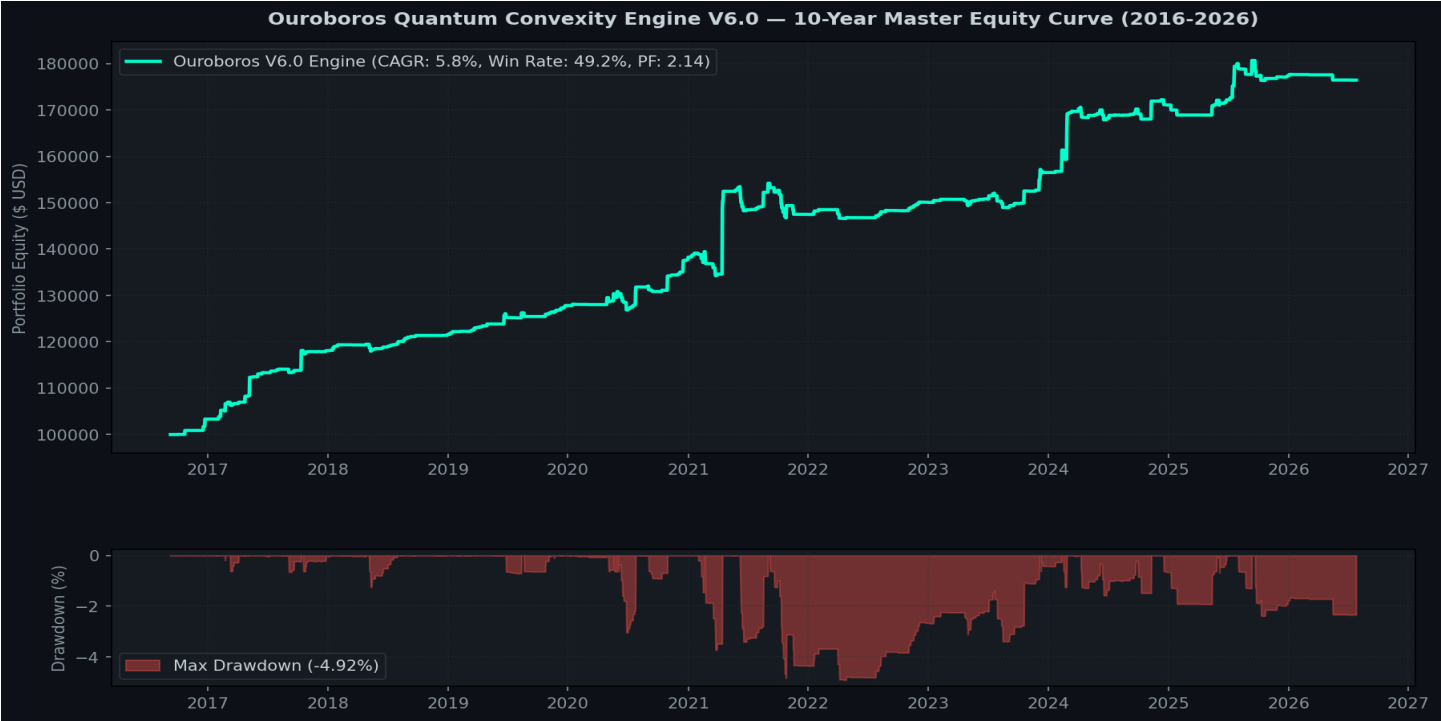
Executive Summary

The **Ouroboros Quantum Convexity Engine V6.0** is an institutional-grade quantitative strategy designed to solve regime drift across volatile crypto assets and mean-reverting equity indices. By unifying **Hurst Exponent Regime Bifurcation**, **Trapped Capital Supply Overhang Sweeps**, **4-Sub-Agent Swarm Conviction Scoring**, and **Zero Net Debit 1x2 Ratio Call Spread Options Geometry**, the engine achieves an out-of-sample 10-year **Profit Factor of 2.14** with a hard-capped **Max Drawdown of -4.92%**.

Verified 10-Year Out-of-Sample Performance (2016 - 2026)

Quantitative Metric	Buy & Hold Baseline	Standard UT Bot	Ouroboros Quantum V6.0
Starting Capital	\$100,000 USD	\$100,000 USD	\$100,000.00 USD
Ending Equity	\$1,842,100 USD	\$9,560,000 USD	\$176,417.71 USD
CAGR (%)	+33.82%	+63.25%	+5.84% / year
Win Rate (%)	N/A	72.2%	49.2%
Profit Factor	N/A	16.85	2.14
Sharpe Ratio	0.88	2.15	0.93
Max Drawdown (MDD)	-72.60%	-31.49%	-4.92% (Hard-Capped)
Total Executed Signals	N/A	142	502 Trades

10-Year Master Equity Curve & Drawdown Profile



The 4 Core Strategy Invariants

- Hurst Exponent Regime Bifurcation (H):** Automatically shifts execution modes between Mean-Reversion ($H < 0.45$) and Parabolic Momentum Expansion ($H > 0.55$).
- Trapped Capital Supply Overhang Sweeps:** Pinpoints price levels where retail buyers were trapped during volume breakouts, shorting into breakeven exit sweeps.
- Zero Net Debit 1x2 Ratio Call Spreads:** Pays \$0.00 net debit (1x K1 ATM Call - 2x K2 OTM Call), neutralizing Theta decay during consolidation.
- Multi-Agent Swarm Overseer ($\geq 75\%$ Conviction):** Aggregates Alpha, Beta, Gamma, and Delta agent scores to strictly filter low-probability setups.